

Abstract No.

Identification and therapeutic targeting of paracrine signaling cues inducing mesenchymal drift of infiltrating glioblastoma

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Glioblastoma (GBM) almost invariably relapse and the recurrent tumor likely derives from invasive tumor cells escaping neurosurgical resection and adjuvant treatment. Interaction of GBM cells with the microenvironment fosters therapy resistance, but we lack detailed knowledge on the protumorigenic signaling cues in the invasive zone. Here, we used an orthotopic mouse model to monitor the transcriptomic states of genetically engineered, human GBM cells and observed that acquisition of a “mesenchymal” (MES) state in the infiltrative margin promotes chemoresistance. Also, we identified parenchymal cells inducing mesenchymalization of invasive GBM cells. This was achieved with engineered tumor cells that labeled neighboring, microenvironmental cells promoting mesenchymal drift in GBM. Using this model, we confirmed a predominant role for myeloid cells in initiating mesenchymalization of the tumor core. However, in the tumor invasive margin other parenchymal cells had higher capacity for the induction of MES-GBM and in particular mesenchymal stromal cells (MSC) were identified as powerful inducers of mesenchymal shifts. In vitro gene expression analysis revealed that MSCs modulate tumor cells through TGF- β induced phosphorylation of SMAD2/3 and increased ID1 expression. In in vivo experiments we investigated the interaction of MSCs with invasive GBM cells. We corroborated that invading GBM cells neighboring with MSCs acquired a MES state, while in other infiltrating GBM cells the proneural subtype dominated. Activity of the pSMAD2/3 and ID1 pathways directly correlated with mesenchymalization of GBM. Combinatorial application of the TGF- β Blocker Galunisertib (GAL) together with temozolomide attenuated pSMAD2/3-and ID1-signaling and tumor mesenchymal drift. Notably, the number of gamma H2A.X positive, infiltrating GBM cells was strongly increased after coapplication of GAL with TMZ (as compared to chemotherapy alone). Altogether, MSCs induced PN-to-MES transition can strongly contribute to therapy resistance and tumor relapse. Systemic administration of Galunisertib blocks this pathological pathway and supports standard clinical care.



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